

P R O J E C T

Q-Catch

Multi-Agent Reinforcement Learning
in Adversarial Gridworld Environments

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Presentation Outline



Problem & Motivation

Multi-agent pursuit–evasion in gridworlds



Theoretical Background

MDPs, Markov Games, Q-Learning foundations



Algorithms

Minimax-Q, Nash-Q, Dyna-Q/Q+, DQN



Scenario 1: Base Chase

5×5 predator–prey pursuit game



Scenario 2: Wumpus Chase

7×7 grid with hazards, treasure, percepts



Experimental Results

Role outcomes and opponent-diversity pilot



Key Findings & Conclusion

Role specialization and limits of generalization

Problem & Motivation

The Challenge

Multi-agent reinforcement learning (MARL) extends single-agent RL to settings where multiple agents interact.

Core difficulty:

The environment is non-stationary from each agent's perspective — dynamics depend on the evolving policies of all other agents.

This violates the stationary MDP assumption and complicates convergence guarantees.

Our Approach

Two adversarial gridworld scenarios:

1. Base Chase — pure pursuit–evasion on a 5×5 grid
2. Wumpus Chase — extended 7×7 grid with hazards, treasure, and perceptual cues

Suite of Q-learning-based methods:

Fictitious Play, Minimax-Q, Nash-Q, Dyna-Q, Dyna-Q+, and DQN — evaluated in self-play and cross-play matchups.

Theoretical Framework

Markov Decision Processes

Tuple (S, A, P, R, γ) — find policy π^* maximizing expected cumulative discounted reward. Bellman optimality equations provide iterative update targets.

Markov Games (Stochastic Games)

Extends MDPs to N agents with joint actions. Each agent i has its own reward R_i . Zero-sum: minimax value. General-sum: Nash equilibrium as solution concept.

Cooperative

All agents share the same reward. Joint policy maximizes shared return.

Competitive (Zero-Sum)

One agent's gain is the other's loss ($R_2 = -R_1$). Minimax value is the solution concept.

General-Sum

Independent rewards, neither fully aligned nor opposed. Nash equilibria as solution concept.

Learning Algorithms

Q-Learning

Model-free tabular RL. Off-policy updates using TD error. Converges to Q^* under Robbins–Monro conditions.

Minimax-Q

Two-player zero-sum extension. Uses LP to compute maximin security policy at each state. Converges to game value.

Nash-Q

General-sum extension. Computes Nash equilibrium of stage game at each state for TD targets. Requires equilibrium selection.

Dyna-Q / Q+

Hybrid model-based + model-free. Learns transition model online; performs N simulated planning updates per step. Q+ adds exploration bonus for unvisited pairs.

DQN

Neural network function approximation for Q-values. Experience replay + target network for stability. Lightweight variant used as baseline.

Scenario 1

Base Chase

Environment Specification

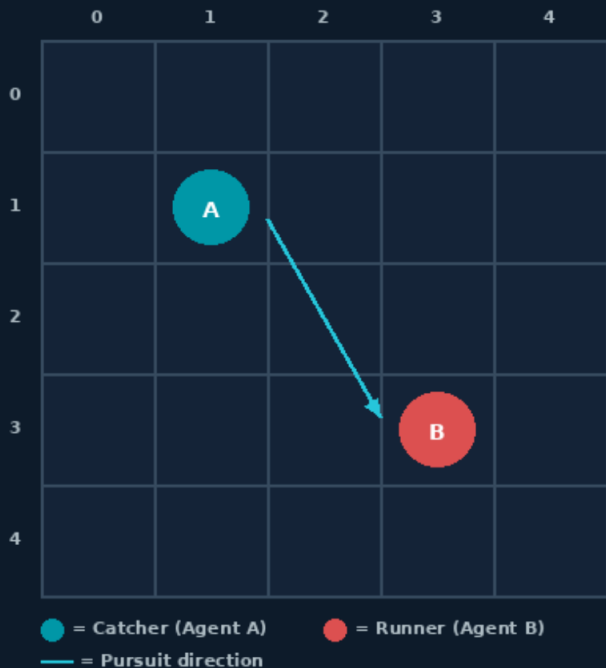
Grid	5 × 5 (625 joint states)
Actions	STAY, UP, DOWN, LEFT, RIGHT
Dynamics	Simultaneous moves, $p_{\text{fail}} = 0.10$
Capture	Same cell or cross-path swap
Termination	Capture or $t_{\text{max}} = 30$

Reward Design

Capture	+10.0 / -10.0
Step penalty	-0.01 (Catcher only)
Wall penalty	-0.02
Move reward	+0.01
Distance shaping	± 0.02

Base Chase

Grid Environment



Pursuit-Evasion Dynamics

Agent A (Catcher) pursues Agent B (Runner) on a 5x5 grid. Both agents move simultaneously. The Catcher aims to minimize distance and capture; the Runner aims to maximize distance and survive until timeout.

Simultaneous Moves + Stochasticity

Actions are executed concurrently. Each action independently fails with probability $p_{\text{fail}} = 0.10$, keeping the agent in place. Moves are clamped to the grid boundary.

Capture Conditions

Capture occurs if agents occupy the same cell after moving (same-cell) or swap positions in one step (cross-path). The episode ends on capture or at $t_{\text{max}} = 30$ steps.

State space: $(x_A, y_A, x_B, y_B) \rightarrow 625$ joint states | Action set: {STAY, UP, DOWN, LEFT, RIGHT}

Scenario 2

Wumpus Chase

Extended Mechanics

Grid	7×7 ($t_{\max} = 40$)
Hazards	Pits (lethal) + Wumpus (lethal)
Percepts	Breeze, Stench, Glitter, Bump
Treasure	Secondary win condition
Roles	A = Chaser (fixed), B = Runner (fixed)

Terminal Priority (No Draws)

1

Capture

Same cell / swap $\rightarrow$ A_WIN

2

Treasure

First to reach $\rightarrow$ winner (tie $\rightarrow$ A)

3

Hazard

B dies $\rightarrow$ A_WIN; A dies $\rightarrow$ B_WIN

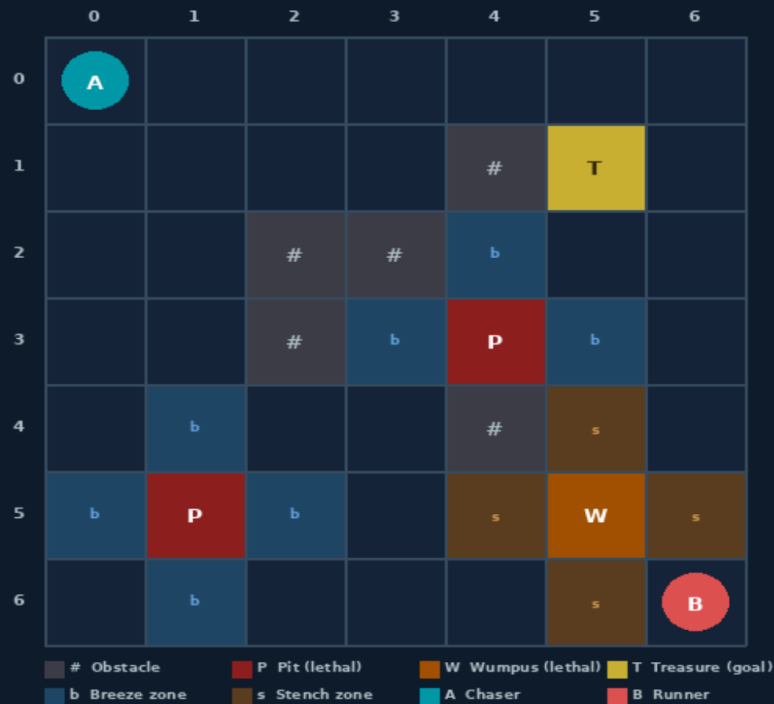
4

Timeout

$t \geq t_{\max} \rightarrow$ B_WIN

Wumpus Chase

Map Layout & Hazards



Lethal Hazards

Pits at (3,4) and (5,1) kill any agent that steps on them. The Wumpus at (5,5) is likewise lethal. Hazard deaths trigger immediate termination with large negative reward for the victim.

Perceptual Cues

Breeze cells (adjacent to pits) and stench cells (adjacent to Wumpus) provide local warning signals. Agents in these zones receive small perception penalties as risk-aware shaping.

Treasure & Competing Pressures

Treasure at (1,5) creates a secondary win path. Agents must balance pursuit/evasion, hazard avoidance, and treasure acquisition. Obstacles at 5 cells constrain movement corridors.

Experimental Setup

Training Protocol

- Episodic multi-agent training over 10k–30k episodes
- ϵ -greedy exploration: 0.20→0.05 (Chase), 0.30→0.05 (Wumpus)
- Simultaneous action execution at each step
- Fixed random seeds for reproducibility

Evaluation Protocol

- Periodic eval every 50 (Chase) / 100 (Wumpus) episodes
- 10 greedy episodes per checkpoint ($\epsilon = 0$)
- Metrics: win rate, avg steps, avg reward
- Offset eval seeds to test generalization

Shared Hyperparameters

Parameter	Chase	Wumpus Chase
Grid size	5 × 5	7 × 7
t_max	30	40
p_fail	0.10	0.10
γ (discount)	0.95	0.95
α (learning rate)	0.10	0.10
Eval frequency	Every 50 ep	Every 100 ep

Chase Results: Performance

Minimax-Q vs Minimax-Q

Strongest overall profile: highest stable capture rate plateau with lowest steps-to-capture. Most probability mass at low step counts.

FP vs FP

Rapid capture rate rise and stabilization. Competitive with Minimax-Q self-play but slightly longer episodes on average.

Minimax-Q vs FP

Near-zero capture rate. Episodes almost always terminate by timeout. Driven by strategic mismatch, not mechanical failure.

Key Observations

Robustness to stochasticity: Increasing p_{fail} from 0.1 to 0.2 does not change qualitative ranking across matchups.

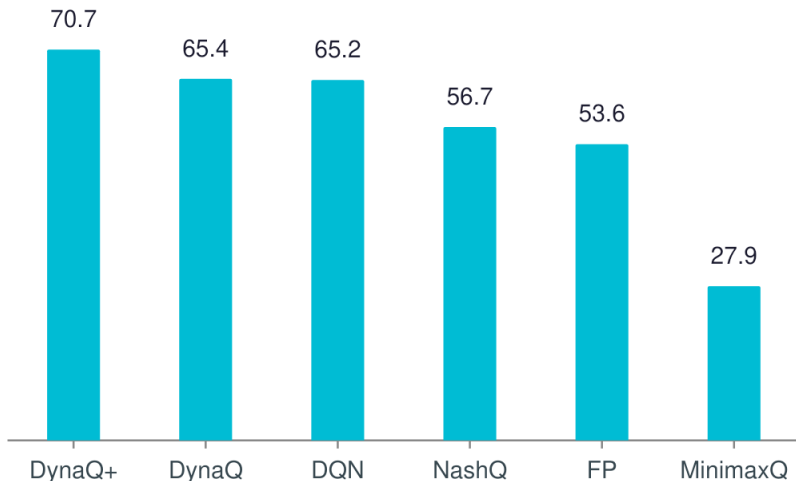
Opening distance control: Rolling opening Manhattan distance stays in a narrow band — performance differences reflect policy quality, not initialization bias.

Failure rates: Runner's observed failure rate is typically higher than Catcher's — policy–environment interaction modulates empirical failure beyond configured p_{fail} .

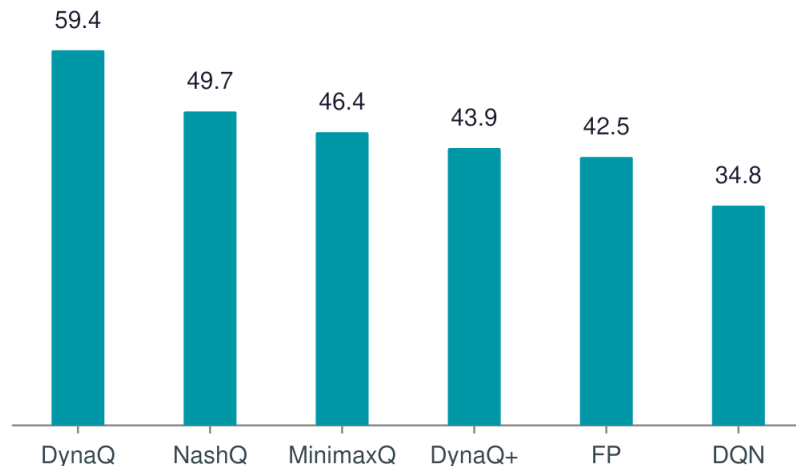
Spatial behavior: Catcher concentrates on central cells ("control-the-center"); Runner favors perimeter/boundary evasion.

Wumpus Chase: Win Rates by Role

Runner Win Rate (%)



Chaser Win Rate (%)



Takeaway

Model-based planning variants dominate: DynaQ is the best Chaser (0.594 win rate, only method with clearly positive terminal reward at 1.814). DynaQ+ is the best Runner (0.707 win rate, highest terminal reward at 4.104). DQN achieves competitive Runner win rate (0.652) but is the weakest Chaser (0.348).

Wumpus Chase: How Episodes End

DynaQ+ Runner

Timeout dominates (~50%)

Survives until late termination. Avg 27.5 steps — longest episodes. Converts survival into wins via timeout rule.

DynaQ Chaser

Chaser treasure ~39%

Wins by reaching treasure, not just capture. Avg 11.3 steps — fastest resolution. Only Chaser with clearly positive terminal reward.

DQN (both roles)

High timeout share (~33%)

Slower, less decisive policies. Competitive Runner win rate but middling avg steps. Lowest mean goal distance yet weak Chaser performance.

Hazards are pit-dominated: ~91.7% of Chaser hazard endings and ~76.6% of Runner hazard endings are caused by pits (not Wumpus)

Wumpus Chase: Behavioral Diagnostics

Average Terminal Reward

Agent	Runner	Chaser
DynaQ+	+4.104	-1.274
DynaQ	+2.992	+1.814
DQN	+3.013	-3.158
NashQ	+1.170	-0.908
FP	-0.197	-3.389

Average Terminal Steps

Agent	Runner	Chaser
DynaQ+	27.46	12.94
DynaQ	15.95	11.34
DQN	16.61	13.62
FP	19.93	15.95
MinimaxQ	15.08	12.64

Behavioral Insights

Action selection: STAY is the most common action (~38% Chaser, ~39% Runner) — rational safety in hazard-rich environment but correlates with timeout-heavy outcomes.

Spatial control under pressure: DynaQ+ Runner relocates to a distinct grid subset when Chaser is within distance ≤ 2 — evidence of mode-switching policy (evasive vs goal-directed).

Favorable distance progress: Best Chasers: DynaQ & NashQ (~0.26). Best Runner: FP (0.341). DQN scores low on both — its win rate relies on alternative termination dynamics, not distance shaping.

Train vs eval gap: Runner opening rate drops in eval (mean $\Delta \approx -0.055$). DQN most sensitive; FP closest to zero.

Extension: Opponent Diversity and Robustness

Does opponent diversity improve performance against unfamiliar strategies?

Dueling Double DQN in every arm 3 seeds 20,000 steps per model

Arm	Training opponent	Action failure probability
A	Fixed FP or Minimax-Q	0.10
B	Uniform opponent pool	0.10
C	Fixed FP or Minimax-Q	0.05 / 0.10 / 0.15
D	Uniform opponent pool	0.05 / 0.10 / 0.15
E	Adaptive family $\times$ noise sampling	0.05 / 0.10 / 0.15

Pilot: 42 models and 252,000 evaluation episodes

Primary test: unseen treasure / ambush opponents at $p_{\text{fail}} = 0.20$ and 0.30 .

Separate 5×5 study with time-aware states and zero-sum potential rewards.
Not directly comparable with the preceding 7×7 benchmark.

Pilot Results: Role-Dependent Generalization

Primary endpoint: win rate against unseen families at $p_{\text{fail}} = 0.20 / 0.30$

DQN training condition	Chaser	Runner
A · fixed FP	45.2%	34.6%
A · fixed Minimax-Q	50.5%	44.0%
B · uniform pool	54.7%	43.2%
C · fixed FP + random p	45.3%	35.3%
C · Minimax-Q + random p	56.0%	49.4%
D · pool + random p	56.8%	44.2%
E · adaptive sampling	54.6%	44.2%

Chaser

D has the highest mean, close to C–Minimax-Q. D 95% CI: 52.4–61.1%.

Runner

C–Minimax-Q leads on mean. Population training does not win against every baseline.

Adaptive sampling adds no clear benefit over D.

D minus mean A controls: Chaser +8.9 pp (95% CI –4.8 to +22.6).
Runner +4.9 pp (95% CI +2.7 to +7.0).

Pilot evidence: 3 seeds, one map and two unseen scripted families. Nominal CIs, no multiplicity correction.

Key Findings



Role Specialization

DynaQ is the strongest Chaser (0.594 win rate, +1.814 avg reward). DynaQ+ is the strongest Runner (0.707 win rate, +4.104 avg reward). Different methods excel at different strategic roles.



Termination Mechanisms Matter

Success often comes through non-obvious paths. DynaQ Chaser wins via treasure (~39%), not just capture. DynaQ+ Runner wins through timeout survival (~50%).



Hazards Are Pit-Dominated

Among hazard endings, pits cause ~92% of Chaser deaths and ~77% of Runner deaths. Pit avoidance is the primary hazard-management factor.



Pilot Generalization

Population training reaches the highest Chaser mean in the pilot. The Runner favors a fixed Minimax-Q opponent with randomized dynamics. Adaptive sampling adds no clear benefit.

Conclusion

The original benchmark shows role specialization and alternative win paths through capture, treasure and survival.

In the separate 5×5 pilot, population training with randomized dynamics reaches the highest Chaser mean. The Runner performs best on mean with a fixed Minimax-Q opponent.

Adaptive sampling adds no clear benefit. Three seeds and two unseen scripted families support preliminary, role-dependent findings rather than a general claim of superiority.



Thank You